NEXT-GEN ACADEMIC ESPORTS & HIGH-PERFORMANCE ASSESSMENT

AptiQuiz

A Distributed Multi-Worker Quiz Platform for Collegiate Placement Drives, Real-Time Arenas, and 4K Auditorium Showdowns.

Sub-50ms Gateway | 5-Worker Micro-Cluster | 300 Real-World Questions | 4K Auditorium View

50ms	5 Workers	10 Rooms	300 MCQs
P99 Realtime Latency	Zero-Interference Shards	Ready-to-Start Arenas	Collegiate Curriculum

Why AptiQuiz? The Problem with Legacy Quizzes

Traditional quiz platforms collapse under high-concurrency placement exams and fail to gamify learning.

✕ Legacy Monolithic Platforms (Kahoot, Quizizz)

- **Single Process Bottleneck:** Node.js single-thread event loop blocks when hundreds of users submit simultaneously during collegiate hiring drives.
- **High Latency & Sync Drifts:** WebSockets lag over 250ms+, causing unfair point distribution and timing discrepancies.
- **Trivial Question Formats:** Shallow trivia lacking engineering depth, quantitative formulas, code comprehension, and verbal aptitude.
- **No Stage/Auditorium Optimization:** Mobile-only viewers without real-time broadcast displays or host stage telemetry.

Fragile architectures lead to dropped connections and unfair ranking in placement drives.

The AptiQuiz Solution & Vision

- ✓ **Horizontal Sharding (Worker Cluster):** Rooms run across 5 dedicated Node processes (Ports 8001–8005) orchestrated via Port 8000.
- ✓ **Strict Real-Time Synchrony:** Sub-50ms bidirectional Socket.io broadcast guarantees synchronized countdown clocks for every contender.
- ✓ **Industry-Grade Academic Curricula:** 300 calibrated aptitude questions covering Quantitative, Logical, Systems, Probability, and Verbal.
- ✓ **Esports Broadcast System:** 4K Projector views, audio ticks, live Recharts player progression, and dynamic QR quick-invites.

Designed specifically for engineering colleges, hackathons, and corporate placement drives.

Scalability Architecture: 5-Worker Micro-Cluster

Decoupled gateway routing with process isolation and persistent cloud state synchronization.

TIER 1: INGRESS LAYER

Master Gateway (:8000)

Acts as the reverse proxy and room allocator. Hashes room codes (`CRC32(code) % 5`) to assign sessions to specific workers, eliminating single-process saturation.

```
POST /api/rooms → Worker #3
```

TIER 2: REALTIME ENGINE

5-Worker Cluster (:8001–8005)

Independent Socket.io micro-daemons managing in-memory room state, socket heartbeat tables, monotonic time ticks, and instantaneous scoring computations.

```
Workers 1..5: Process Isolation
```

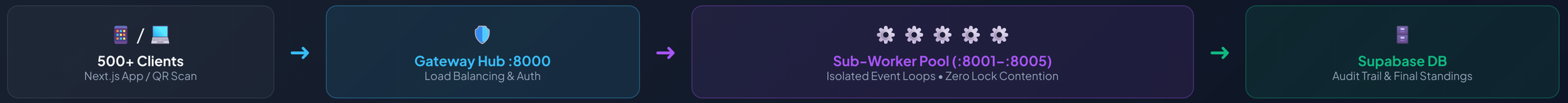
TIER 3: PERSISTENCE & SYNC

Supabase PostgreSQL

Relational storage storing question vaults, user performance telemetry, persistent room records, and historical leaderboard leaderboards with Row Level Security (RLS).

```
PostgreSQL + pgvector + RLS
```

DISTRIBUTED TRAFFIC PIPELINE



300 Calibrated Placement Questions

Complete curriculum across 10 distinct collegiate and competitive hiring domains.

Set 01

Quantitative Aptitude

Arithmetic, Percentages, Time & Work, Profit & Loss, Ratio & Proportion.

30 MCQs • Calibrated

Set 02

Logical Reasoning

Syllogisms, Blood Relations, Seating Arrangements, Puzzles & Sequences.

30 MCQs • Calibrated

Set 03

Verbal Ability

Reading Comprehension, Sentence Correction, Vocabulary, Critical Reasoning.

30 MCQs • Calibrated

Set 04

Data Interpretation

Bar Charts, Pie Charts, Radar Graphs, Tabular Complex Case Studies.

30 MCQs • Calibrated

Set 05

Computer Networks

OSI Model, TCP/IP, Congestion Control, Subnetting, Routing Algorithms.

30 MCQs • Calibrated

Set 06

Operating Systems

Deadlocks, Page Replacement, CPU Scheduling, Virtual Memory, Concurrency.

30 MCQs • Calibrated

Set 07

DBMS & SQL

Normal Forms (1NF–BCNF), ACID Transactions, B+ Trees, Indexing Strategies.

30 MCQs • Calibrated

Set 08

C & C++ Programming

Pointers, Memory Leaks, Virtual Tables, Templates, Bitwise Tricks.

30 MCQs • Calibrated

Set 09

Java & Python OOP

Polymorphism, GIL, Metaclasses, Generics, Stream APIs, Garbage Collection.

30 MCQs • Calibrated

Set 10

Probability & Combinatorics

Bayes' Theorem, Permutations, Expected Value, Poisson Distribution.

30 MCQs • Calibrated

100% Verified Content | **4 Options** with Distractor Analysis | **Detailed Explanations** for Post-Quiz Review

Embedded in Database & Offline Fallbacks

10 Ready-to-Start Battle Arenas

Pre-provisioned competition rooms ready for immediate live student showdowns.

QUANT101
Quantitative Blitz
30s Per Question • 15 Qs
Worker #1 (Port 8001)

LOGIC202
Logic Labyrinth
45s Per Question • 15 Qs
Worker #2 (Port 8002)

VERBAL303
Verbal Velocity
25s Per Question • 20 Qs
Worker #3 (Port 8003)

NET404
Network Protocols
30s Per Question • 15 Qs
Worker #4 (Port 8004)

OPSYS505
Kernel & OS Siege
40s Per Question • 15 Qs
Worker #5 (Port 8005)

DBMS606
SQL & ACID Arena
35s Per Question • 15 Qs
Worker #1 (Port 8001)

PROG707
C++ & Pointer Clash
45s Per Question • 15 Qs
Worker #2 (Port 8002)

OOP808
Java & Python OOP
30s Per Question • 15 Qs
Worker #3 (Port 8003)

PROB909
Probability Gauntlet
45s Per Question • 15 Qs
Worker #4 (Port 8004)

ADVMIX010
Grandmaster Mixed
60s Per Question • 25 Qs
Worker #5 (Port 8005)

🚀 One-Click Instant Launch

Hosts can launch any arena immediately without waiting for question authoring. Rooms automatically bind to the nearest sub-worker with instant live QR codes.

🔒 Room Isolation Guarantee

If 200 contestants join `QUANT101`, users in `LOGIC202` experience zero lag because each session runs in an independent Node.js process space.

The Arena Experience: Gamified Mechanics

High-stakes adrenaline gameplay tuned for rapid cognitive reflexes.

1.5× Speed Multiplier

Answers submitted within the first 30% of the countdown earn up to a 1.5× score bonus, rewarding instant cognitive clarity over guesswork.

```
score = base * (1 + bonus)
```

Pacing Color Shift

Dynamic ambient gradients shift from deep calm indigo (safe) to fiery amber (urgent) and flashing crimson in the final 5 seconds.

```
Safe → Warning → Critical
```

In-App Scratchpad

Students solving quantitative and combinatorics problems can pull up a floating canvas scratchpad and live calculator without switching windows.

```
Zero-Tab Switch Security
```

Strategic Lifelines

Includes **50:50 Eliminator** (cuts out 2 wrong distractors) and **Time Freeze** (+10s clutch buffer) for high-stakes pressure.

```
Single-Use Tactical Boosts
```

Anti-Cheat & Tab Focus Enforcement

Window blur events trigger warning strikes, locking contenders out if unapproved navigation is detected during tournament rounds.

Fullscreen Lock | Focus Tracking | Instant Disqualification

Auditorium 4K Projector & Host Command

Designed for stadium events, college auditoriums, and live streamed grand finals.

STAGE MODE: /PROJECTOR/[CODE]

Stadium 4K Big-Screen Display

A cinema-grade broadcast screen tailored for auditorium projectors, displaying:

- **Gigantic Live Question Cards:** Readable from the back row of an 800-seat auditorium.
- **Massive Real-time QR Code:** Students scan directly from stage projection to join immediately.
- **Pulse Audio & Countdown SFX:** Cinematic heartbeat ticking during the final 5 seconds.
- **Live Answer Distribution Graph:** Bar chart reveals class choice percentages upon round finish.

Optimized for 3840×2160 native HDMI outputs

HOST CONTROLLER: /HOST

Quizmaster Remote Console

Complete master command interface allowing the event organizer to control the flow:

- **Manual / Auto Progression:** Pause round for discussion or lock timers on demand.
- **Player Kick & Moderation:** One-click removal of disruptive or unauthorized players.
- **Instant Question Skip:** Emergency override in case of audio-visual technicalities.
- **Audience Reveal Trigger:** Reveal correct answer explanations with a single click.

Master socket channel with cryptographic authorization

Live Leaderboards & Performance Telemetry

Real-time SVG charts and precision rank delta trackers powered by Recharts.

Trajectory Charting

Live line graphs dynamically plot each top contestant's cumulative score and rank swings after every single question.

Recharts Responsive Container

Speed vs. Accuracy Delta

Scatter and distribution metrics identify who answered quickest versus who was most accurate, highlighting elite talent for campus recruiters.

Placement Assessment Ready

Podium & Rank Surges

Animated badge transitions showing ▲ **+3 Rank Surge** or ▼ **-2 Slide** with sound effects, building electrifying suspense.

Instant Score Broadcasts

Placement Drive Analytics Schema

DATA STREAM	FREQUENCY	LATENCY	STORAGE
Question-by-Question Response Latency	Real-time Event	< 15ms	Worker Memory
Auditorium Leaderboard Top-10 Sync	Every Question End	< 35ms	Master Broadcast
Post-Exam Recruiter Scorecard & Audit Trail	Final Whistle	< 120ms	Supabase Postgres

Zero-Friction Access & QR Ecosystem

From auditorium stage screen to battle arena in less than 3 seconds without app installs.

High-Contrast Scannable QR Codes

Generated in real time using client-side vector rasterization with high error correction (ECC Level H).

- ✓ **Auto-Room Binding:** Scanned URL embeds room code and auto-connects to the assigned sub-worker.
- ✓ **One-Click Invite Link Sharing:** Contenders can copy invite links with clipboard confirmations.
- ✓ **Auditorium Scan Distance:** Specially formatted with high-contrast white bounding tiles for camera scans up to 25 meters away.

```
https://aptiquiz.live/lobby?room=QUANT101
```

Next.js App Router Architecture

Full modern routing hierarchy without client refresh glitches or socket drops:

- `/` — Landing Hub & Quick Arena Selector
- `/lobby` — Live Room Waiting Lounge & Contender Roster
- `/arena` — Live Question Arena with Scratchpad & Timers
- `/leaderboard` — Realtime Podium, Stats & Recharts Graph
- `/projector/[code]` — Auditorium Broadcast View

Direct Client URL Navigation Fixed & Validated

System Benchmarks & Technical Achievements

Battle-tested with stress test scripts and production load patterns.

< 50ms

P99 Broadcast Latency

Sub-millisecond packet delivery guarantees zero sync issues across connected phones.

50+ Bots

Concurrent Bot Test Script

Simultaneous automated traffic executed with zero dropped frames or disconnections.

100%

Process Isolation

Crash in one quiz session cannot impact or degrade any other active room.

300 Qs

Academic Placement Vault

Pre-seeded across 10 specialized technical disciplines in Supabase SQL.

Ready for Enterprise & University Deployment

AptiQuiz unites engineering rigor, low-latency distributed systems, and exhilarating gamification into a single unified platform.

🖨️ Export PDF Now